

Manuscript title:

Restrictions on fronting and stress shift in the encoding of Czech focus

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Abstract: This article examines two operations which are used to encode focus in Czech: word order alternation and stress shift. Based on data from a previous production experiment, we ran a rating study where participants rated the authenticity of utterances with manipulated information structure. The results showed that alternating word order to encode focus is free, supporting the claims about the flexibility of Czech word order. Shifting stress is a more costly operation – it is possible (in terms of competence), but it is not necessarily easy (in terms of performance). Applying both word order alternation and stress shift leads to a significant decrease in rating. We argue that this decrease is caused by a penalty on an uneconomical structure when there is an economical one at hand which respects default word order and default prosody; and not by the simple cumulativeness of these two operations.

keywords: focus · Czech · word order · stress shift · information structure

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1 Introduction

Cross-linguistically, information structural concepts such as focus or givenness have been observed to be encoded by several means: word order alternation, prosodic marking, morphology, particles, or syntactic constructions such as clefts. In this article we zoom in on the first two encoding strategies in Czech, a language which has been claimed to exhibit both flexible word order and flexible prosody. This is illustrated by example (1), from Groeben et al. (2017, 186), who supplied evidence from an acceptability rating experiment that subject focus may be encoded equally well by stress shift (from the default final position to the marked initial position), see (1-B₁), and by changing the word order (from the default SVO to the marked OVS), see (1-B₂).¹

- (1) A: ‘Who drank alcohol on that day?’
- B₁: [Josef NĚMEC]_F pil ten den alkohol.
Josef Němec drank DEM day alcohol
- B₂: Alkohol pil ten den [Josef NĚMEC]_F.
alcohol drank DEM day Josef Němec
- For both: ‘Josef Němec drank alcohol on that day.’

The more nuanced experiments of Šimík & Wierzba (2017) have shown that both stress shift and word order alternations (aka scrambling) come with a certain cost (reflected in an acceptability decrease), whereby the former appears to be a bit more costly than the latter. This can be illustrated by example (2) (from Šimík & Wierzba 2017, 688), where the need to destress the discourse given object is achieved more naturally by using the noncanonical SOV order, see (2-B₁), than by shifting the stress to the verb, see (2-B₂). Still, both are more acceptable than any condition with the object stressed, showing the strength of the requirement to destress given elements.

- (2) A: ‘We’d like to talk to the Italian ambassador.’
- B₁: Myslím, že pan Sládek [toho velvyslance]_G KONTAKTOVAL.
think that Mr Sládek DEM ambassador contacted
- B₂: Myslím, že pan Sládek KONTAKTOVAL [toho velvyslance]_G.
think that Mr Sládek contacted DEM ambassador
- For both: ‘I think that Mr. Sládek has contacted the ambassador.’

Šimík & Wierzba (2017) argue that the differences in acceptability do not reflect grammaticality distinctions, but rather preferences. They also assume that constraint violations, e.g. the deviation from the canonical stress or word order pattern, contribute cumulatively to

¹The word that is set in SMALL CAPS is the locus of the sentence stress. The subscripts F and G indicate focus and givenness, respectively.

the overall acceptability decrease. That is, if stress shift causes a penalty x and scrambling a penalty y , the overall penalty assigned to a clause where both operations have applied would be $x + y$. The authors notice that this cumulativity assumption does not always seem to be sufficient. More specifically, the acceptability decrease caused by shifting the stress to a scrambled focused object, as in (3) (from Šimík & Wierzba 2017, 688; our judgement added), is considerably more dramatic than what one would expect by a simple cumulation of the penalties caused by shifting the stress and by scrambling.

- (3) A: ‘Do you have an idea who Mr. Sládek contacted in the end?’
 B: ?Myslím, že pan Sládek [toho VELVYSLANCE]_F kontaktoval.
 think that Mr Sládek DEM ambassador contacted
 Intended: ‘I think that Mr. Sládek has contacted the ambassador.’

Šimík & Wierzba (2017) hypothesize that the unacceptability of (3) could reflect a specific constraint on the scrambling of focused elements to a non-peripheral position, known from literature on German (aka A-scrambling); see Lenerz (1977). We note that, intuitively, the scrambling of a focused object to the clausal periphery (aka A-bar scrambling) is more acceptable:

- (4) A: ‘Do you have an idea who Mr. Sládek contacted in the end?’
 B: Myslím, že [toho VELVYSLANCE]_F pan Sládek kontaktoval.
 think that DEM ambassador Mr Sládek contacted
 ‘I think that Mr. Sládek has contacted the ambassador.’

In this paper, we compare the acceptability of stress shift to focused subjects, as in (1), with the acceptability of stress shift to fronted (A-bar-scrambled) focused objects, as in (4). Assuming that there is a qualitative difference between A- and A-bar scrambling, we consider two hypotheses: (i) Stress shift to a fronted object will exhibit a cumulative penalty of stress shift and fronting (following the spirit of Šimík & Wierzba’s 2017 proposal). (ii) The penalty associated with a stress shift to a fronted object will be bigger than what one would expect from a cumulative effect of two constraint violations; the rationale behind this hypothesis (following the spirit of Reinhart 2006) is that fronting a focused object involves two costly operations in a situation where neither is in principle required: the same effect can be achieved by using a canonical stress pattern (final stress) and canonical word order (SVO). Our experimental results support hypothesis (ii).

In addition, we will show that while stress shift is generally available in Czech grammar, it is not easy to produce for less proficient native speakers in an experimental setting. We interpret this as an indication of the relative strength of the requirement to place sentence stress clause-finally (aka nuclear stress rule; see below), already observed in Šimík & Wierzba (2017).

1 The article is organized as follows. In section 2 we summarize previous claims about
 2 information structure in Czech and its encoding in word order and prosody as well as
 3 introduce the relevant constraints that govern linguistic form, especially word order and
 4 sentence prosody. Section 3 presents the rating experiment which was designed to compare
 5 the effects of the constraints on the perceived naturalness of utterances. We describe the
 6 procedure, design and materials as well as the statistics used to analyze the results. We
 7 discuss the findings in section 4.

8 2 Background

9 Czech word order is often said to be free (Siewierska & Uhlřová, 1998). As opposed to
 10 English for example, utterances may exhibit various orderings of the subject, verb and
 11 object. However, it has often been claimed that the order of Czech words is conditioned by
 12 certain categories of information structure (= IS).² For example, Kučerová (2011) proposes
 13 that in Czech given constituents must precede new ones and analyzes this phenomenon as
 14 an effect of a specific operator called G. A number of observations has been made about
 15 the syntactic marking of focus. Traditionally, it has been claimed that focus in Czech
 16 tends to be placed utterance-finally (Mathesius, 1939; Sgall et al., 1980; Firbas, 1992).
 17 The role of word order has been predominant in the discussions of IS encoding in Czech so
 18 far. It should be noted that the previous accounts use different definitions of focus (calling
 19 it also rheme or rheme proper) and givenness, either based on the notion of importance,
 20 highlighting or contextual boundness, which differs from our approach described below.

21 Similarly to word order, prosody in Czech is claimed to be flexible. The role of prosody
 22 in Czech IS encoding has been emphasized by e.g. Mathesius (1939) or Daneš (1957). As
 23 for focus, they both notice that although a focused constituent can occur in any position in
 24 the utterance, it requires prosodic prominence. In this way Czech focus behaves similarly
 25 to other Slavic languages (for Russian, see, Alter e.g. 1997a,b; Zybatow & Mehlhorn e.g.
 26 2000; for Polish, Hamlaoui et al. 2018; for a recent survey, see Šimík 2024).

27 In the present article, we consider IS notions as binary oppositions, based on the
 28 account in Krifka (2008). Namely, we work with the oppositions of focus-background
 29 and given-new. A focused constituent gives rise to alternative denotations relevant in the
 30 current discourse (Rooth, 1992); background is what is not focused. A given constituent
 31 has a synonymic or hyponymic antecedent in the immediately preceding discourse; new
 32 material is what is not given.

33 That focus bears prosodic prominence has been observed before (Jackendoff, 1972;

²Apart from the IS factors, other regularities have been observed, e.g. the fixed placing of clitics in the second position (Wackernagel, 1892).

¹ Truckenbrodt, 1995; Zubizarreta, 1998; Reinhart, 2006).³ This is usually captured by
² the Stress Focus (SF) constraint, provided in (5). Recently there has been a number of
³ empirical studies where the prosodic nature of focus was tested (Baumann et al., 2006;
⁴ Breen et al., 2010; Kügler & Gollrad, 2015, a.o.), for Czech Šimík & Wierzba (2017);
⁵ Hamlaoui et al. (2018). Givenness, on the other hand, leads to (relative) deaccentuation,
⁶ as captured by the constraint in (6). This tendency was also tested empirically, e.g. by
⁷ Baumann & Riester (2013). It is important to note that *SG affects *sentence*, not *lexical*
⁸ stress, and that this constraint does not apply in cases of given elements in narrow focus.

⁹ (5) Stress Focus (SF): Sentence stress is realized in the focus of the sentence.

¹⁰ (6) *Stress Given (*SG): Sentence stress is not realized in a given element.

¹¹ In addition to constraints regulating the relation between information structure and
¹² prosody, Czech utterances show two tendencies towards the default. These tendencies
¹³ can again be formulated as OT constraints (following Šimík & Wierzba 2017). The first
¹⁴ tendency is towards the canonical word order – SVO in declarative sentences – and is
¹⁵ formulated as a constraint in (7). CWO is violated when other word order variants than
¹⁶ SVO are used.

¹⁷ (7) Canonical Word Order (CWO): The linear order of S, V, and O is SVO.

¹⁸ The second tendency is towards default (sentence) prosody. This is captured in (8) by
¹⁹ the Nuclear Stress Rule (NSR) (going back to Chomsky & Halle 1968 in the formal
²⁰ linguistic tradition; for Czech, this was noticed already in Daneš 1957).⁴ It requires that
²¹ the rightmost phrasal stress is assigned extra prominence – it carries the so-called nuclear
²² stress.

²³ (8) Nuclear Stress Rule (NSR): Sentence stress is realized on the rightmost element of
²⁴ the sentence.

²⁵ The NSR constraint can be violated by an operation called Stress Shift (SS). Stress shift
²⁶ can be employed to stress a nonfinal focused constituent. The stress thus leaves its default
²⁷ position and moves leftwards.⁵ Importantly, the two variants (rightmost vs. shifted stress)
²⁸ do not differ in meaning and IS, only in their prosodic realization:

³The focus-as-prominence has been challenged by the focus-as-alignment approach of Féry (2013).

⁴In this article, we make a distinction between stress and prominence. While prominence is a term connected to the prosodic realization, stress is its theoretical counterpart. Prominence is a broad term encompassing higher fundamental frequency, higher intensity, longer duration and specific intonational contours/tones.

⁵Even though we use the term “stress shift” and say that stress “moves”, it is not our intention to propose an analysis of how this happens. What is rather intended is to highlight that this operation overrides the default NSR.

- 1 (9) A: ‘Who ate my ice cream?’
 2 B₁: Myslím, že ti zmrzlinu snědla [JARKA]_F. NSR satisfied, CWO violated
 think that you ice-cream ate Jarka
 3 B₂: Myslím, že [JARKA]_F ti snědla zmrzlinu. NSR violated, CWO satisfied
 think that Jarka you ate ice-cream
 4 For both: ‘I think that Jarka ate your ice cream.’

5 In the present study, the constraints Stress Focus, *Stress Given, Canonical Word Order,
 6 and Nuclear Stress Rule can be thought of as Optimality-Theoretic (OT) constraints
 7 within a weighted (rather than categorical) OT paradigm (Pater 2009; Šimík & Wierzba
 8 2017; see also Keller 2000). This paradigm is well-suited for the modelling of gradient
 9 acceptability of multiple grammatical structures with a comparable pragmatic function.
 10 Each constraint is assigned a non-binary language-specific weight, which corresponds to
 11 the “importance” of satisfying the constraint. If a language has strict word order, for
 12 instance, it has a large weight of Canonical Word Order. A violation of a constraint
 13 correlates, by assumption, with an acceptability decrease: the larger the weight, the bigger
 14 the decrease upon violation. Our hypothesis (i) assumes simple cumulativeness of violations.
 15 Our hypothesis (ii), on the other hand, builds in an additional economy component, which
 16 induces an extra penalty for violating constraints in a situation where the violations could
 17 have been eschewed.

18 3 Experiment

19 We designed a perception rating experiment to test the previously mentioned constraints
 20 (CWO and NSR) and their possible violations in Czech.

21 In the following section we describe the procedure and participants, then we explain
 22 the experimental design and comment on the materials used. Finally, we present the
 23 results and discuss them.

24 3.1 Procedure and participants

25 The present experiment builds on a previous production study where we elicited and
 26 recorded utterances from 68 Czech native speakers. The stimuli were in the form of
 27 constructed dialogues, which consisted of a cue and a target. The cue utterances were
 28 pre-recorded by professional actors (a female and a male), while the target utterances by
 29 the 68 native speakers. An example stimulus (dialogue) is shown in (10).

- 30 (10) A: Kdo na té oslavě s Kateřinou tančil?
 who at DEM party with Kateřina danced
 31 ‘Who danced with Kateřina at the party?’

1 B: Myslím, že [David]_F tančil s Kateřinou.
 think that David danced with Kateřina
 2 'I think that David danced with Kateřina.'

3 For the purposes of the present rating study we used a subset of the recordings from the
 4 production study, which contained the desired manipulations of CWO and NSR violations.
 5 The participants were presented with the audio stimuli, and their task was to rate the
 6 authenticity of the target utterance, i.e. the second speaker's reaction to the first speaker.⁶
 7 To do this, participants selected a value on a Likert scale ranging from 1 (completely
 8 inauthentic) to 7 (completely authentic). Each participant rated either 94 or 82 targets in
 9 a randomized order, depending on the experimental list they were assigned. In total we
 10 had 228 participants. All of them were self-identified native Czech speakers and students
 11 of Charles University. They were compensated with ECTS credits for their participation.
 12 Before the task began, participants read instructions, agreed with the participation and
 13 tried out the rating on test stimuli. The experiment was run online on the platform
 14 PsyToolkit (Stoet, 2010, 2017).

15 We chose authenticity as the dependent variable for two reasons. Firstly, one of the
 16 goals of this study was to see how proficient our speakers from the production experiment
 17 were. We expected that their perceived authenticity as speakers would reflect their
 18 proficiency in IS encoding. The mean authenticity scores of the 68 speakers are shown in
 19 Fig. 1. Later on we will make a distinction between more proficient speakers (the top ten
 20 rated ones) and the less proficient ones (the rest), thus arriving at two speaker groups.

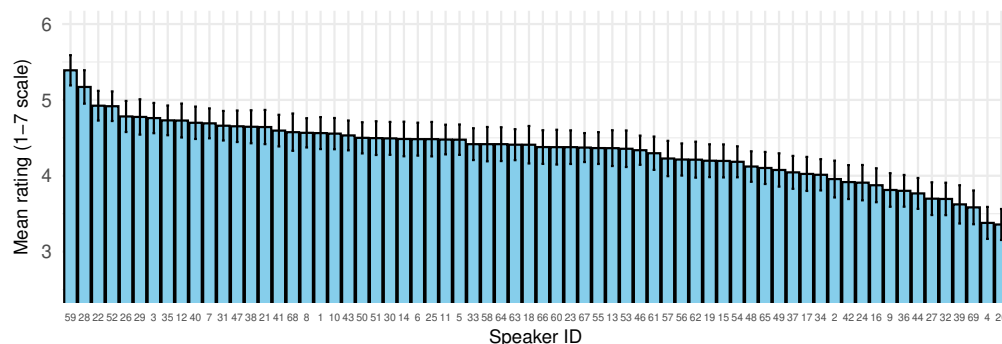


Figure 1: Mean ratings of speaker authenticity on a scale from 1 to 7 (on the y axis) for each of the 68 speakers from the production experiment (on the x axis).

21 Secondly, we assume that authenticity as a variable correlates with the acceptabil-
 22 ity/naturalness of an utterance. This means that whether an utterance is perceived as more
 23 or less authentic depends not only on the speaker but also on the variables manipulated

⁶The participants were instructed that the cues (A) were recorded by professional actors and the responses (B) by non-professionals attempting their best possible performance.

1 in the design. More specifically, we assumed that the authenticity of the target response
 2 was influenced by the violation of CWO, NSR, or both together.

3 3.2 Design & materials

4 The design of the main experiment was a within-participant and within-item 2×2 design.
 5 We manipulated two categorical variables: SYNTACTIC FUNCTION of the focused constituent,
 6 which was either the subject or the object, and PRESENCE OF STRESS SHIFT, which was
 7 there or not. Table 1 schematically summarizes the four conditions. The expectations
 8 (Section 3.3) and statistical analysis will be formulated in terms of the satisfaction (or
 9 violation) of the two constraints under investigation – CWO and NSR, as also detailed
 10 in Table 1. CWO and NSR will thus be used as categorical (binary) predictors in the
 11 statistical models.

condition	schema	SYNT. FUNCTION	STRESS SHIFT	CWO	NSR
a	SV[O] _F	obj	no	+	+
b	[S] _F VO	subj	yes	+	–
c	OV[S] _F	subj	no	–	+
d	[O] _F VS	obj	yes	–	–

Table 1: Schematic design of the experiment; + means constraint satisfied; – constraint violated

12 Both manipulations were present in the target utterance in each dialogue, prompted
 13 by the utterance of the first speaker, as illustrated in (11). Each target contained one
 14 focused constituent (subject or object) and the background to it was always contextually
 15 given. We included focused constituents used as an answer to a *wh*-question (shown in
 16 (10)) as well as corrective foci (shown in (11)).

- 17 (11) A₁: Vypadá to, že tam servírka čepuje limonádu. for cond. a + d
 seems it that there waitress draws lemonade
 ‘It seems that that waitress there has lemonade on tap.’
 18
 19 A₂: Vypadá to, že támhleten mladík čepuje pivo. for cond. b + c
 seems it that that.DEM boy draws beer
 ‘It seems that that young guy has beer on tap.’
 20
 21 B: a. Řekl bych, že servírka čepuje [pivo]_F.
 22 b. Řekl bych, že [servírka]_F čepuje pivo.
 23 c. Řekl bych, že pivo čepuje [servírka]_F.
 24 d. Řekl bych, že [pivo]_F čepuje servírka.
 said AUX that beer draws waitress
 25 For all: ‘I’d say that the waitress has beer on tap.’

26 Our four conditions correspond to a violation pattern concerning CWO and NSR. Condition

(a) does not violate any of these two, since it has the canonical SVO word order and the stress is placed finally (i.e. it is not shifted). Condition (b) does not violate CWO, but it violates NSR by shifting the stress from the final constituent onto the subject in the initial position. Condition (c) does not violate NSR, but violates CWO. Finally, condition (d) violates both CWO and NSR as the stress shifts onto the initial object in the OVS word order.

The stimuli were distributed into 20 experimental lists. Each list contained utterances by 8 or 9 different target speakers, to ensure variability of the stimuli. The order of stimuli in each list was randomized. We made sure that a participant saw only one condition per item. The stimuli from the main experiment were interspersed with fillers – the filler dialogues had the same structure as the main ones, but did not involve the crucial manipulation of the independent variables described above.

3.3 Expectations

The expectations we formulate are based on the assumption that constraint violations correlate with perceived naturalness/acceptability and, in our specific case, with authenticity. If, for instance, the NSR is violated (if stress is shifted), we expect this to have a negative effect on authenticity rating; call it a rating drop. The magnitude of the effect correlates with the weight of the constraint violated. In a language with a small weight of CWO, for instance, noncanonical word orders will be perceived as highly acceptable.

Regarding the two hypotheses formulated in Section 2, they both predict rating drops upon CWO/NSR violations. Both predict the highest rating for condition (a), where no constraint is violated, and the lowest rating for condition (d), where CWO and NSR are violated at the same time. The difference between them is that hypothesis (i) predicts the drop in condition (d) to be the result of a cumulative violation of both CWO and NSR, while hypothesis (ii) predicts a more dramatic rating drop, caused by the fact that the same information structure could have been conveyed without any violation at all, as in condition (a). In terms of our statistical analysis, hypothesis (i) predicts the presence of main effects of both CWO and NSR, but no interaction between the two; hypothesis (ii) also predicts an interaction.

3.4 Results

The results were analyzed in RStudio using the package `tidyverse` Wickham et al. (2023). Statistical tests were run in RStudio using a Cumulative-Link Mixed Model (function `clmm` of the package `ordinal`; Christensen 2022) and `emmeans` (package `emmeans`; Lenth 2025). A Cumulative-Link Mixed Model is suitable for ordered categorical data (in our case this

1 was the rating on the scale from 1 to 7). It uses the maximum likelihood estimation (MLE)
 2 which then allows for model comparison. In the model, our random effects were stimulus,
 3 participant (rater) and speaker (the voice of the target utterance), to account for their
 4 variability. The emmeans method counts the estimated marginal mean for each condition.
 5 The R code with the models and their results is available on OSF.⁷

6 The raw results of authenticity rating for the four conditions are shown in Fig. 2. The
 7 data from the total of 68 speakers were split into two groups based on their perceived
 8 proficiency: the more proficient speakers are the top rated, the less proficient are the rest.

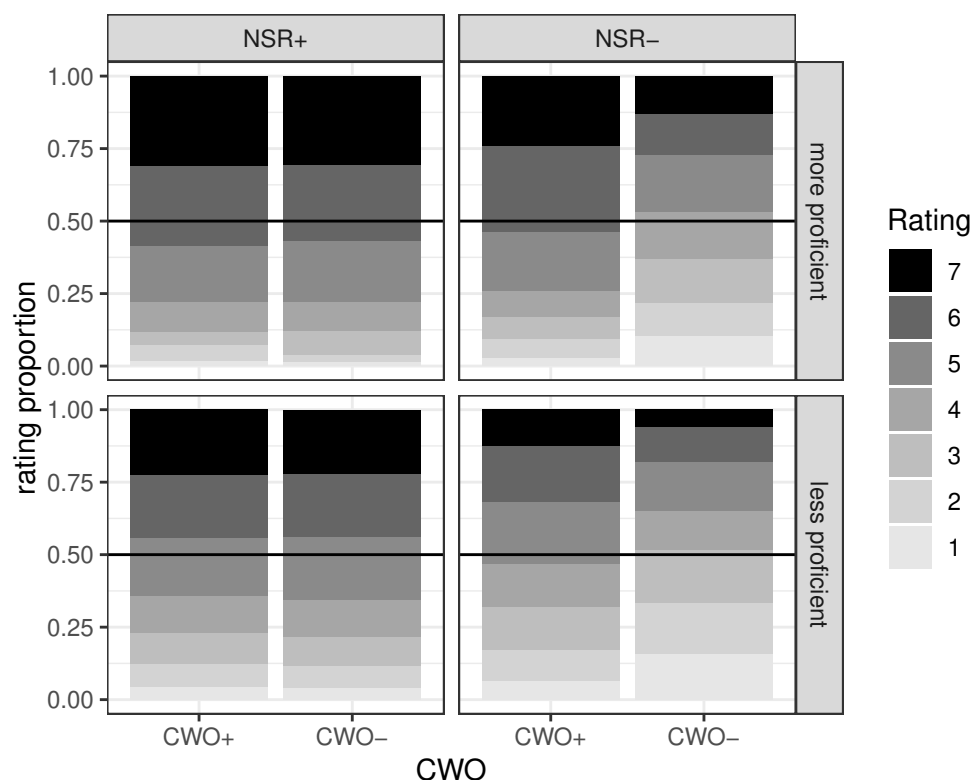


Figure 2: Ratings of the four conditions for speakers divided by their proficiency – raw results. The y-axis shows the proportion of ratings. The points of the 1-7 Likert scale are represented by the shades. The horizontal line cuts through the median for each condition.

9 For this data we fitted a Cumulative-Link Mixed Model predicting the value of rating
 10 from CWO, NSR, SPEAKER PROFICIENCY and their interactions. All of these factors
 11 had two levels: CWO+ and CWO– for CWO, NSR+ and NSR– for NSR, and more
 12 proficient vs. less proficient for SPEAKER PROFICIENCY. All were treatment coded, with
 13 CWO+, NSR+, and less proficient as reference levels, respectively. We included random
 14 slopes for participants, stimuli and speakers. The resulting fixed effects (coefficients)

⁷The link to the repository is here: https://osf.io/yfm8q/?view_only=759195cad0914ca28cf63dd562e9f2ac.

	Estimate	Std. Error	z value	p value	
CWO	0.032 34	0.124 08	0.261	0.79	
NSR	−0.589 91	0.113 92	−5.178	<0.001	***
PROFICIENCY	0.853 71	0.241 30	3.538	<0.001	***
CWO × NSR	−1.038 54	0.070 17	−14.800	<0.001	***
CWO × PROFICIENCY	−0.077 30	0.193 77	−0.399	0.69	
NSR × PROFICIENCY	0.011 73	0.263 70	0.044	0.97	
CWO × NSR × PROFICIENCY	0.049 99	0.214 70	0.233	0.82	

Table 2: Fixed effects (coefficients) of the Cumulative Link Mixed Model fitted

are presented in Table 2. The interaction between CWO and NSR, visible in Table 2 was confirmed to be significant (at $p < .001$) also by comparing the model with another model excluding the relevant interaction term, using the function `anova()`.⁸ The interaction can be characterized as follows: the drop in rating caused by an NSR violation is more pronounced if CWO is violated at the same time. Next, the model reveals a main effect of PROFICIENCY ($p < .001$): responses of more proficient speakers are rated as more authentic than responses of less proficient speakers. Finally, the model shows a main effect of NSR ($p < .001$): utterances with shifted stress are rated as less authentic than utterances with default stress. A post-hoc test, based on computing the estimated marginal means (the results are visualized these results in Figure 3), shows that the main effect of NSR is largely contingent on the aforementioned interaction between NSR and CWO.

4 Discussion

The presence of a two-way interaction between CWO and NSR – the joint violation of CWO and NSR (condition (d)) exhibiting a rating drop that goes beyond the plain cumulation of the individual CWO/NSR violations (conditions (b) and (c)) – is consistent with the expectations drawn from hypothesis (ii). This implies that Czech speakers are not only sensitive to the markedness of noncanonical order and noncanonical stress (hypothesis (i)), but also that they implicitly contrast the uneconomical option of stress shift to a fronted constituent ([O]_FVS) with the more economical default order and stress (SV[O]_F), assuming that both convey the same meaning and information structure. Furthermore, the result is similar to what Šimík & Wierzba (2017) found for non-peripheral (A-scrambled) foci. Whether the reduced acceptability (authenticity) of our [O]_FVS condition and Šimík & Wierzba’s S[O]_FV condition (see example (3)) has a common denominator, or not, cannot be decided based on the data available so far: a single experiment comparing the

⁸This method is called the Likelihood Ratio Test (LRT), which compares a complex model with a simpler one in order to determine whether the added complexity of the former significantly improves the fit to the data compared to the latter.

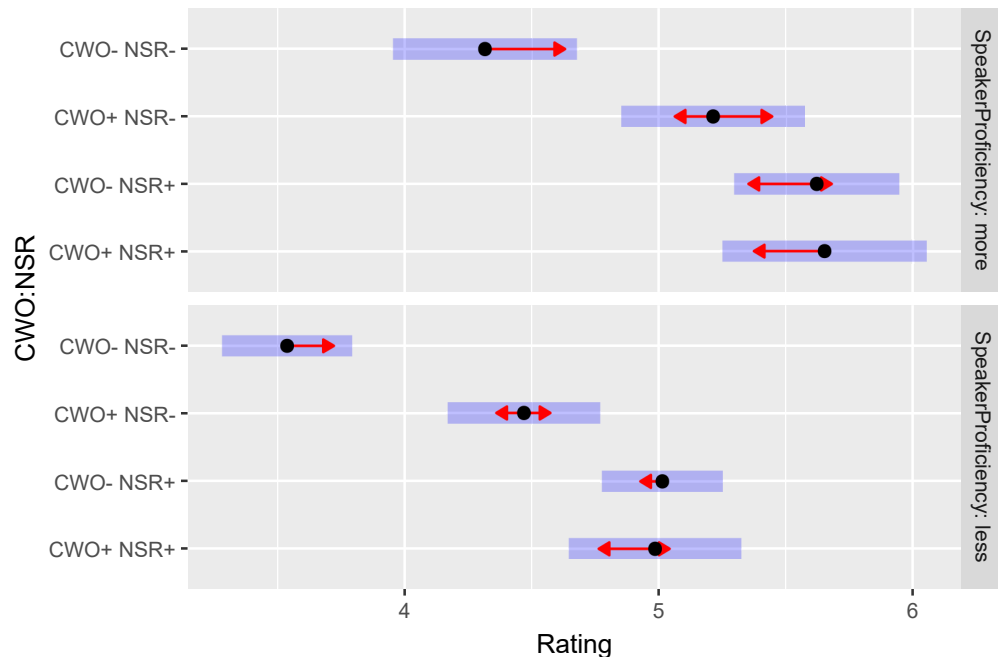


Figure 3: Ratings of the four conditions for speakers divided by their proficiency – emmeans results. The x axis shows the EMMs of ratings. The blue bars are confidence intervals for the EMMs, and the red arrows are for the comparisons among them. Whenever the red arrows overlap between two conditions, it means that there is no statistically significant difference between the two conditions.

two conditions would be needed. Our own intuition – expressed by the contrast between (3) and (4) – is that $[O]_F$ VS is more natural.⁹ One question that remains open is which factor would make the critical $[O]_F$ VS (or $[O]_F$ SV) condition fully acceptable. Our results show that focus itself is not sufficient. We remind the reader that our items were based on two uses of focus: answerhood and corrective. However, an auxiliary analysis revealed no effect of focus use: both exhibited the same pattern. Among the factors to be considered in future work are exhaustivity (aka exhaustive identification; É. Kiss 1998; Horvath 2010), mirativity/unexpectedness (Cruschina, 2011; Trotzke, 2017), and register (Wierzba & Fanselow, 2020; Marklová et al., 2024).¹⁰

Let us now move on to the individual effects of NSR and CWO. Our results reveal a main effect of NSR – stress shift is less authentic than default stress – but no effect of CWO – noncanonical order is as authentic as the canonical one. This finding, which is consistent with that reported in Šimík & Wierzba (2017), suggests that while both word

⁹What there is evidence for is that there seems to be no distinction between $[O]_F$ VS and $[O]_F$ SV (see Kořánová to appear). This is consistent with Šimík & Wierzba’s (2017) finding that word order within the background/given/unstressed domain has no clear effect on acceptability.

¹⁰In this context, it should be mentioned that already Mathesius (1947) noted that clause-initial focus (which he referred to as “subjective word order”) is associated with evaluative or mirative implications. Nevertheless, as far as we understand his suggestion concerned all clause-initial foci irrespective of whether they are fronted or not.

order and prosody are flexible in Czech, the former is more flexible than the latter, or, in other words, that the weight of NSR is larger than that of CWO.¹¹ That said, the numerical rating difference between NSR-only violations (condition [S]_FVO) and CWO-only violations (condition OV[S]_F) is small compared to what we observe for NSR+CWO violation (condition [O]_FVS). This implies that there is no economy consideration applied when comparing NSR-only violations with CWO-only violations; in other words, the capacity of a language to change word order does not block the availability of stress shift.

Besides the expected main effect of proficiency (more proficient speakers are rated as more authentic than less proficient ones), post-hoc pairwise comparisons of individual conditions (visualized in Figure 3) within the more proficient and less proficient subsets reveal a notable pattern related to the authenticity of NSR-violating conditions, or, in other words, the naturalness of stress shift. The results of the more proficient group replicate the findings reported in Groeben et al. (2017), namely that stress shift to subjects (condition [S]_FVO) is as acceptable as placing subjects clause-finally (condition OV[S]_F) (see example (1) above).¹² However, these two conditions differ significantly in the less proficient subset: the former (stress shift) condition is rated as less authentic than the latter (noncanonical order) condition. We take this proficiency-based difference to indicate that stress shift in Czech is in principle free, but it is not easy to produce in an experimental setting. If less proficient speaker fail to apply stress shift adequately – whether due to insufficient prominence on the initial constituent or due to insufficient destressing of the background – it leads to the violation of an information structural constraint (Stress Focus, *Stress Given, or both), and therefore to an additional rating drop not found in the more proficient group.

5 Conclusion

In the present article we examined the role of word order alternations and stress shift in focus encoding in Czech. Both of these operations arguably come with a cost, although stress shift seems to be more costly (Šimík & Wierzba, 2017). We compared stress shift to focused subjects vs. stress shift to fronted focused objects and expected that stress shift to a fronted object would correlate with a decrease in acceptability. The decrease was hypothesized to be a result of either a cumulative penalty on stress shift and fronting, or a penalty on using two costly operations in a situation where neither is necessary from

¹¹In Šimík & Wierzba (2017), the estimated weights of NSR and CWO in their most successful “combined model” was -0.41 and -0.33 , respectively. The estimated weights correspond to the rating drop measured in standard deviations.

¹²We note that the result is replicated despite the fact that Groeben et al. (2017) had matrix clause stimuli, while we had embedded clause stimuli. Whether the matrix vs. embedded contrast plays any role for object focus fronting is a question for future research.

the perspective of information structure.

To test the hypotheses, we designed a rating experiment which was based on a previous production study. In the production study, 68 Czech native speakers read utterances (targets) as reactions to pre-recorded cues. In the targets, information structure as well as word order were manipulated. Selected recordings from the production experiment were used as stimuli in the present rating experiment, where participants rated the authenticity of the target utterances. The targets in the selected stimuli featured the manipulation of the syntactic function of the focused constituent (subject vs. object) and of the presence of stress shift (present vs. not present). The resulting 4 conditions differed in the constraint violations they involved. Of particular interest was the condition $[O]_F$ VS, which received the lowest rating. Our results suggested that this decrease in rating was caused by the categorical – rather than cumulative – effect of violating two constraints (CWO and NSR). We conclude that mere constraint cumulation is not sufficient in accounting for the acceptability of focused object fronting in Czech; instead, the reduced acceptability is a result of a global comparison between candidates: the costly $[O]_F$ VS structure is penalized because the free SV $[O]_F$ could have been used.

Additionally, our data showed that shifting stress in Czech is possible (in terms of the system/competence), but not necessarily easy (in terms of performance). The latter point was demonstrated on the ratings of the more proficient vs. less proficient speakers, where the less proficient group had difficulties in producing the expected stress shift. This underscores the conclusion that while both word order and prosody are flexible in Czech (Vallduví & Engdahl, 2013; Šimík & Wierzba, 2017), the former is more flexible than the latter.

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